EXERCISE 4

Damped Forced Oscillator: Amplitude and Phase of the Steady State

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Frequency Response • Amplitude M • Phase φ

STATEMENT

Consider the oscillating system of the previous exercise, but now in the presence of a non-negligible viscous drag so that Newton's second law reads

$$\ddot{x} + \lambda \dot{x} + \omega^2 x = a \cos \alpha t,$$

in the underdamped case in which $\lambda^2 < 4\omega^2$. Show that $x_p(t) = M \cos(\alpha t - \phi)$ is a particular solution, finding its amplitude M and phase ϕ in terms of the system parameters. Discuss how the relative ratio between the system response and the external forcing, M/a, behaves with the external frequency α and compare with the zero-drag ($\lambda = 0$) case.

PHASE 1: SETUP AND HYPOTHESES

Parameters: $\lambda > 0$ (damping rate, s^{-1}), $\omega > 0$ (natural frequency), $\alpha > 0$ (forcing frequency), $a > 0$ (forcing amplitude per unit mass). Underdamped: $\lambda^2 < 4\omega^2$, so the characteristic roots are $-\frac{\lambda}{2} \pm i\sigma$ with $\sigma = \frac{1}{2}\sqrt{4\omega^2 - \lambda^2} > 0$ (Topic 3, Sec. 4.3) and the complementary function is $x_h = e^{-\lambda t/2}(c_1 \cos \sigma t + c_2 \sin \sigma t)$, which decays. Only x_p survives at long times.

PHASE 2: FORMULATION: WHAT HAS TO BE SHOWN

We must find constants M and ϕ such that $x_p = M\cos(\alpha t - \phi)$ satisfies $L[x_p] = a\cos\alpha t$. Because $\cos(\alpha t - \phi)$ is a shifted cosine, the cosine and sine terms of the guess of Topic 3, Sec. 6.1 are packed in one amplitude and one phase. The subtraction formulas $\cos(\alpha t - \phi) = \cos\alpha t \cos\phi + \sin\alpha t \sin\phi$ and $\sin(\alpha t - \phi) = \sin\alpha t \cos\phi - \cos\alpha t \sin\phi$ will be used.

PHASE 3: DERIVATION

Substitution

Differentiate with the chain rule ($\frac{d}{dt}(\alpha t - \phi) = \alpha$):

$$\dot{x}_p = -M\alpha \sin(\alpha t - \phi), \qquad \ddot{x}_p = -M\alpha^2 \cos(\alpha t - \phi)$$

Therefore

$$L[x_p] = \ddot{x}_p + \lambda \dot{x}_p + \omega^2 x_p = M\left[(\omega^2 - \alpha^2)\cos(\alpha t - \phi) - \lambda\alpha \sin(\alpha t - \phi)\right]$$

Expand the shifted functions and collect cosines and sines of αt:

$$L[x_p] = M\left[(\omega^2 - \alpha^2)\cos\phi + \lambda\alpha \sin\phi\right]\cos\alpha t + M\left[(\omega^2 - \alpha^2)\sin\phi - \lambda\alpha\cos\phi\right]\sin\alpha t$$

This must equal $a\cos\alpha t + 0 \cdot \sin\alpha t$ for every t; $\cos\alpha t$ and $\sin\alpha t$ are linearly independent, so both coefficients must match:

$$M\left[(\omega^2 - \alpha^2)\cos\phi + \lambda\alpha\sin\phi\right] = a \quad (1)$$
$$M\left[(\omega^2 - \alpha^2)\sin\phi - \lambda\alpha\cos\phi\right] = 0 \quad (2)$$

Phase

$M \neq 0$, because $M = 0$ would turn equation (1) into $0 = a > 0$. Hence equation (2) gives $(\omega^2 - \alpha^2)\sin\phi = \lambda\alpha\cos\phi$, i.e.

$$\tan\phi = \frac{\lambda\alpha}{\omega^2 - \alpha^2}$$

To fix the quadrant, introduce $R = \sqrt{(\omega^2 - \alpha^2)^2 + \lambda^2\alpha^2} > 0$ and the angle ϕ defined by

$$\cos\phi = \frac{\omega^2 - \alpha^2}{R}, \qquad \sin\phi = \frac{\lambda\alpha}{R}$$

(these satisfy $\cos^2\phi + \sin^2\phi = \frac{(\omega^2-\alpha^2)^2 + \lambda^2\alpha^2}{R^2} = 1$ and equation (2): $(\omega^2 - \alpha^2)\frac{\lambda\alpha}{R} - \lambda\alpha\frac{\omega^2-\alpha^2}{R} = 0$).
Because $\sin\phi = \lambda\alpha/R > 0$, $\phi \in (0, \pi)$: $\phi = \operatorname{arccot}\frac{\omega^2-\alpha^2}{\lambda\alpha}$, i.e. $\phi = \arctan\frac{\lambda\alpha}{\omega^2-\alpha^2}$ for $\alpha < \omega$ and $\phi = \pi + \arctan\frac{\lambda\alpha}{\omega^2-\alpha^2}$ for $\alpha > \omega$.

Amplitude

Substitute $\cos\phi$ and $\sin\phi$ in equation (1):

$$M\left[(\omega^2-\alpha^2)\frac{\omega^2-\alpha^2}{R}+\lambda\alpha\frac{\lambda\alpha}{R}\right]=M\,\frac{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}{R}=M\,R=a$$

$$\boxed{M=\frac{a}{\sqrt{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}},\qquad \phi=\operatorname{atan2}\big(\lambda\alpha,\ \omega^2-\alpha^2\big)}$$

so $x_p=M\cos(\alpha t-\phi)$ is indeed a particular solution. It agrees with the sine-cosine form of Topic 3, Eq. 15.5: $M\cos\phi=\frac{a(\omega^2-\alpha^2)}{R^2}$ and $M\sin\phi=\frac{a\lambda\alpha}{R^2}$ are exactly the coefficients D and C found there.

Ratio of response to forcing

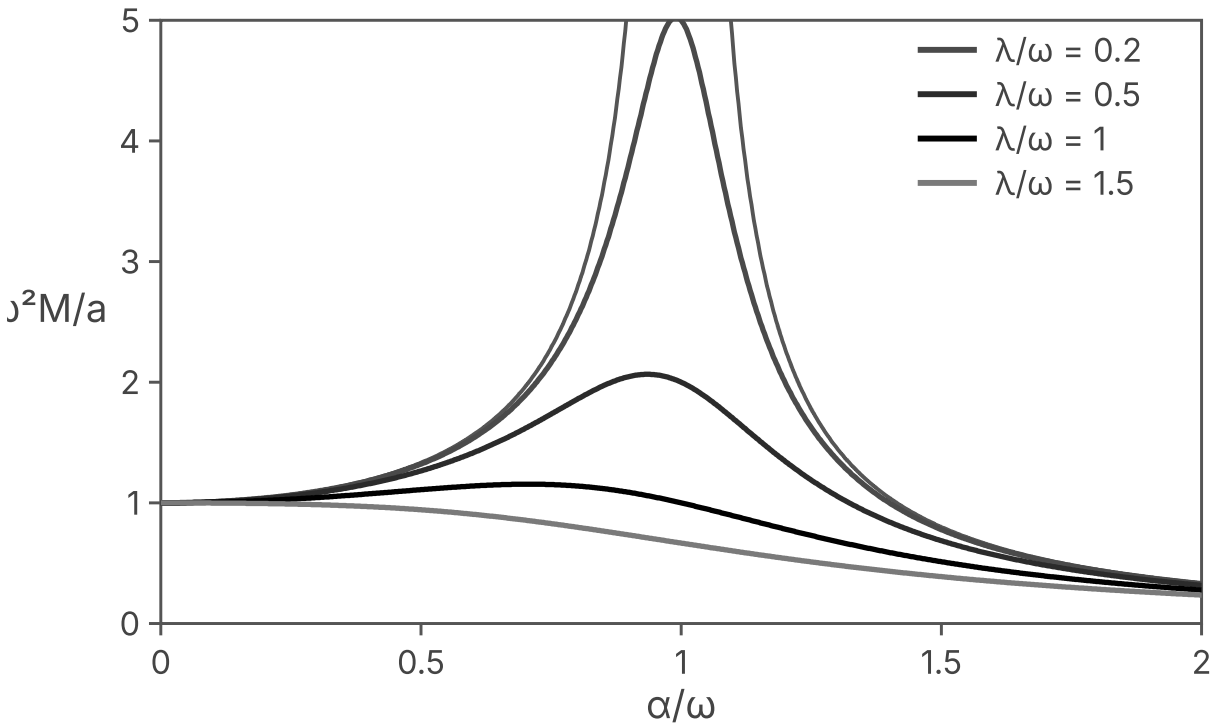
$$\frac{M}{a}=\frac{1}{\sqrt{(\omega^2-\alpha^2)^2+\lambda^2\alpha^2}}=:F(\alpha)$$

Differentiate the radicand $D(\alpha)=(\omega^2-\alpha^2)^2+\lambda^2\alpha^2$ with the chain rule: $D'(\alpha)=-4\alpha(\omega^2-\alpha^2)+2\lambda^2\alpha=2\alpha\left(2\alpha^2-2\omega^2+\lambda^2\right)$. The critical points are $\alpha=0$ and $\alpha_r^2=\omega^2-\frac{\lambda^2}{2}$, which is real only if $\lambda^2<2\omega^2$.

- **Case $\lambda^2<2\omega^2$ (light damping).** D has a minimum at $\alpha_r=\sqrt{\omega^2-\lambda^2/2}<\omega$ and F a maximum. There $\omega^2-\alpha_r^2=\frac{\lambda^2}{2}$, so $D(\alpha_r)=\frac{\lambda^4}{4}+\lambda^2\left(\omega^2-\frac{\lambda^2}{2}\right)=\lambda^2\left(\omega^2-\frac{\lambda^2}{4}\right)$ and $\left(\frac{M}{a}\right)_{\max}=\frac{1}{\lambda\sqrt{\omega^2-\lambda^2/4}}=\frac{1}{\lambda\sigma}$: finite, large for small λ, and attained slightly below ω (pseudo-resonance).
- **Case $2\omega^2\le\lambda^2<4\omega^2$.** $D'(\alpha)\ge0$ for $\alpha>0$: no peak, and M/a decreases monotonically from its value at $\alpha=0$.
- **Limits.** $\alpha\to0$: $M/a\to1/\omega^2$ (static deflection under a constant force); $\alpha=\omega$: $M/a=1/(\lambda\omega)$; $\alpha\to\infty$: $M/a\approx1/\alpha^2\to0$.
- **Phase.** ϕ increases continuously from 0 at $\alpha=0$ through $\pi/2$ at $\alpha=\omega$ (there $\cos\phi=0$) to π as $\alpha\to\infty$: the response lags the forcing by a quarter period exactly at the natural frequency.

Comparison with the undamped case

For $\lambda=0$, $M/a=\frac{1}{|\omega^2-\alpha^2|}$, which tends to $+\infty$ as $\alpha\to\omega$ (resonance, Exercise 3), and ϕ jumps discontinuously from 0 (for $\alpha<\omega$) to π (for $\alpha>\omega$): the response is in phase below ω and in antiphase above. Any $\lambda>0$ regularises this: the peak becomes finite, $1/(\lambda\sigma)$, and the phase jump is smoothed over a band of width $\sim\lambda$ around ω. As $\lambda\to0$, $\alpha_r\to\omega$ and the peak height $1/(\lambda\sigma)\to\infty$, recovering resonance.



Amplitude ratio $\omega^2 M/a$ against α/ω for $\lambda/\omega = 0.2, 0.5, 1, 1.5$ (blue to pink) and the undamped limit (dashed, $\lambda = 0$).

PHASE 4: INTERPRETATION AND CHECKS

RESULT

$x_p(t) = M\cos(\alpha t - \phi)$ with $M = \dfrac{a}{\sqrt{(\omega^2 - \alpha^2)^2 + \lambda^2\alpha^2}}$ and $\phi = \operatorname{atan2}(\lambda\alpha, \omega^2 - \alpha^2) \in (0, \pi)$.

The general solution is $x = e^{-\lambda t/2}(c_1\cos\sigma t + c_2\sin\sigma t) + M\cos(\alpha t - \phi)$: the transient decays and the steady state at the forcing frequency remains, independent of the initial data.

- Numerical check of (1) and (2) at $\omega = 2, \lambda = 1, \alpha = 1, a = 1$: $R = \sqrt{9+1} = \sqrt{10}$, $M = 1/\sqrt{10}$, $\cos\phi = 3/\sqrt{10}$, $\sin\phi = 1/\sqrt{10}$, and $M[3\cos\phi + 1\cdot\sin\phi] = \frac{1}{\sqrt{10}}\cdot\frac{9+1}{\sqrt{10}} = 1 = a$ (checked symbolically in `scripts/verify_topic3_topic4.py`).
- Dimensions: $[\omega^2 - \alpha^2] = [\lambda\alpha] = \mathrm{s}^{-2}$, so $[R] = \mathrm{s}^{-2}$ and $[M] = [a]/[R] = \mathrm{m}/\mathrm{s}^2\cdot\mathrm{s}^2 = \mathrm{m}$; ϕ is dimensionless.
- Condition $\lambda^2 < 4\omega^2$ is only needed for the form of x_h; the formulas for M and ϕ hold for every $\lambda > 0$.